5.3 Cyber security continued

Candidates should be able to:	Notes and guidance
2 Explain how a range of solutions are used to help keep data safe from security threats	• Including: – access levels – anti-malware, including anti-virus and anti-spyware – authentication (username and password, biometrics, two-step verification) – automating software updates – checking the spelling and tone of communications – checking the URL attached to a link – firewalls – privacy settings – proxy-servers – secure socket layer (SSL) security protocol

6 Automated and emerging technologies

6.1 Automated systems

Candidates should be able to:	Notes and guidance
1 Describe how sensors, microprocessors and actuators can be used in collaboration to create automated systems	
2 Describe the advantages and disadvantages of an automated system used for a given scenario	• Including scenarios from: – industry – transport – agriculture – weather – gaming – lighting – science

6.2 Robotics

Candidates should be able to:	Notes and guidance
1 Understand what is meant by robotics	• Robotics is a branch of computer science that incorporates the design, construction and operation of robots • Examples include factory equipment, domestic robots and drones
2 Describe the characteristics of a robot	• Including: – a mechanical structure or framework – electrical components, such as sensors, microprocessors and actuators – programmable
3 Understand the roles that robots can perform and describe the advantages and disadvantages of their use	• Robots can be used in areas including: – industry – transport – agriculture – medicine – domestic settings – entertainment

6.3 Artificial intelligence

Candidates should be able to:	Notes and guidance
1 Understand what is meant by artificial intelligence (AI)	• AI is a branch of computer science dealing with the simulation of intelligent behaviours by computers
2 Describe the main characteristics of AI as the collection of data and the rules for using that data, the ability to reason, and it can include the ability to learn and adapt	
3 Explain the basic operation and components of AI systems to simulate intelligent behaviour	• Limited to: – expert systems – machine learning • Expert systems have a knowledge base, a rule base, an inference engine and an interface • Machine learning is when a program has the ability to automatically adapt its own processes and/or data